

PowerTools for Autodesk Fusion

PowerTools is a single Autodesk Fusion add-in that puts the missing everyday commands back into Fusion, for people who run distributed assemblies, manage team data in Fusion Team, and want the batch, numbering, export and sharing tools that desktop CAD and PDM systems take for granted. One install adds 52 commands to the places you already work (the Design toolbar, the Quick Access Toolbar, the File and Share menus, the Drawing and Animation workspaces), each switchable from one Preferences page.

- **Assemblies:** plan a hierarchy on a canvas and generate the components, externalize local components, infer constraints on an imported STEP, update a distributed design bottom-up.
- **Team data:** hub-unique part and drawing numbers, shared global parameters, where-used and version history views, one-click refresh and close-all.
- **Modeling:** repair and round sketches, flatten doubly-curved faces with a strain map, measure a chain of edges, mirror a part into an opposite-hand document, see which features cost compute time.
- **Exports and sharing:** BOM, Mermaid and SysML exports from the File menu; share, desktop and Fusion Team links from the Share menu.

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Prerequisites

- **Autodesk Fusion** on **Windows** or **macOS**, with add-ins enabled.
- A **Fusion Team** hub for the commands that read or write cloud data (references, numbering, related data, team add-ins, sharing).
- The **Fusion Manage Extension** for Sync Item to Part Number only.

Installation

1. Download or clone this repository.
2. In Fusion, open **Utilities** › **Add-Ins** (or press **Shift+S**).
3. On the **Add-Ins** tab, select the green +.
4. Select the **PowerTools** folder and select **Open**.
5. Select **PowerTools** in the list and select **Run**. Tick **Run on Startup** to load it with Fusion.

Preferences

File › **PowerTools Preferences** on the Quick Access Toolbar opens a palette with one section per command group: an enable checkbox, summary and guide link per command, and settings beneath the commands that have them. Enabling or disabling a command applies on the next Fusion restart; settings apply immediately. **Open settings file** and **Import settings...** back up or restore every choice.

- **Off by default:** Component Warning, Show In Location, Get and Update, Radial Hole Circle, Version Diff.
- **Beta**, hidden until **Show beta commands** is ticked under **General:** Infer Constraints, Radial Hole Circle, Flatten Surface.
- The **Related Data** section sets up the templates folder; the **Team Add-ins** section shows the shared add-ins folder and its settings.

Commands

Most commands live on the **Utilities** tab of the Design workspace, in the **Power Tools** panel. The **Location** column gives the others. Select a command for its guide.

Assembly

Command	Location	Description
Assembly Builder	Design › Utilities › Power Tools	Design an assembly hierarchy in a visual node editor, then generate every external component with the right design intent in one step. Imports a SysML v2 physical view.
Assembly Palette	Design › Assembly › Insert	Quick-start palette for a new assembly: create components in place, insert from thumbnails of open and recent documents, hand off to Assembly Builder, Global Parameters and Fasteners.
Insert STEP File	Design › Assembly › Insert	Insert a STEP file from disk as a component of the active design, with no upload first.
Assembly Statistics	Design › Utilities › Power Tools	Component counts, nesting depth and joint counts for the active design.
Get and Update	Quick Access Toolbar	Get all latest versions and update all assembly contexts in one click. Off by default.
Bottom-up Update	Design › Utilities › Power Tools	Open, update, rebuild and save every referenced document from the leaves up, with a resumable log.
Externalize	Design › Utilities › Power Tools	Save local components as their own cloud documents and re-insert them in place. Converts an imported assembly into a distributed design.
Global Parameters	Design › Utilities › Power Tools	Create and edit parameter sets shared across a project.
Link Global Parameters	Design › Utilities › Power Tools	Derive a shared parameter set into the active design.
Refresh Global Parameters Cache	Design › Utilities › Power Tools	Rescan the project's _Global Parameters folder and rewrite the local cache.
Infer Constraints	Design › Utilities › Power Tools	Propose concentric and coincident constraints for an already-positioned assembly and apply the ones you pick. Beta.
Component Warning	PowerTools Preferences (background)	Warn before a feature is created in the root component or against another component. Off by default.
Change Cycle Color	Right-click menu	Choose the color Component Color Cycling uses for the selected components, instead of the next random one.
Document References	Design › Utilities › Power Tools; Drawing and electronics › Power Tools	Where-used for the active design: root assemblies, parents, children, drawings, fasteners and related data . For a drawing, the designs it documents; for electronics, the rest of the project, schematic, 2D and 3D PCB and drawings.
Refresh Active Document	File	Check the hub for a newer version of the open document and reload it.

Document Tools

Command	Location	Description
Document Information	Design › Utilities › Power Tools; Drawing and electronics › Power Tools	Hub, project, folder, version and MFGDM identifiers of the active design, drawing or electronics document, with a warning when saving would migrate it to a newer Fusion build.
History	Quick Access Toolbar	The document's version history as day rows: a track per author, saves on a clock axis, elapsed time between days.
Version Diff	Design › Utilities › Power Tools	Compare the timeline of two versions of the active design in an HTML report. Off by default.

Command	Location	Description
Assign Part Numbers	Design › Utilities › Power Tools	Hub-unique sequential part numbers (PRT, ASY, WLD, COT, TOL) for the design and its local components.
Assign Drawing Number	Drawing › Power Tools	Hub-unique DWG numbers for drawings, synced into the source design's Drawing Number property.
Sync Item to Part Number	Design › Manage › Power Tools	Copy the Fusion Manage Item Number into the Part Number (Manage Extension).
PowerTools Add Project Folders Favorites	File	Create a standard folder set in the active project, from editable Basic and Advanced lists.
	Quick Access Toolbar	Save folder locations per hub and jump the Data Panel to any of them.
Open Recent	File › Open Recent	Reopen a recent document from the File menu, with location and thumbnail on hover.
Match Units	Design › Inspect	Compare the document's units with your Fusion defaults and change them in one click; the icon shows the state. Optional prompts on open and on entering Manufacture.
Show In Location	PowerTools Preferences (background)	Reveal the document's folder in the Data Panel when a document opens or is activated. Off by default.
Close All Documents	File	Close every open document, saving or discarding unsaved changes as a group.
Toggle Data	Navigation Toolbar	Open or close the Data Panel with one click.
Local Recovery Save	File	Write a local recovery checkpoint without creating a cloud version.

Exports

Command	Location	Description
Export BOM as CSV	File	Flat bill of materials of the assembly as CSV, without making a drawing.
Export Mermaid Diagram	File	The assembly hierarchy as a Mermaid flowchart file, opened in Mermaid Live.
Export SysML Architecture Document	File	The assembly as a 4+1 architecture document with a SysML v2 physical view.

Part Modeling

Command	Location	Description
Round Sketch Dimensions	Sketch › Modify	Round the sketch's length and angle dimensions to a chosen increment, with live preview.
Sketch Under-constrained	Sketch › Modify	Highlight the sketch objects that are not fully constrained.
Radial Hole Circle	Sketch › Create	A construction circle on a picked centre with a diameter dimension and a point at twelve o'clock. Beta, off by default.
Create Mirrored Design	Design › Solid › Create	Derive the design into a new -mirror document and scale it by -1 : an opposite-hand part that follows the source.
Hide Objects	Design › Utilities › Utility	Hide origins, construction geometry, joints, sketches and canvases in every component at once.
Timeline Compute Report	Design › Solid › Inspect	Compute time per timeline feature in an HTML report, with the source CSV.
Measure Path	Design › Inspect	Total length of a connected chain of edges and sketch curves between two picks.
Flatten Surface	Design › Utilities › Power Tools	Flatten curved faces to a sketch pattern with a stretch/gather strain map. Beta.

Animation

Command	Location	Description
Save Named View	Animation › Power Tools	Save the Animation viewport camera as a named view on the design, named from the storyboard.

Related Data

Command	Location	Description
Create Related Data	Design › Solid › Create	Create a Manufacture, Render or other document from a hub template, referencing the active design.
Select Related Data Folder	PowerTools Preferences › Related Data	Choose the hub folder that holds the related-data templates, once per hub.

Team Add-ins

Share add-ins across a team through one hub folder, **Assets / Shared Addins**. Drop an add-in .zip in; everyone else picks it up shortly after Fusion starts, with no restart and nothing to publish.

Command	Location	Description
Team Add-ins	Design › Utilities › Power Tools	Check the shared folder now and show what the last check did.
Set Up Shared Add-ins Folder	PowerTools Preferences › Team Add-ins	Find or create the shared folder in the hub's Assets project.

Tools

Command	Location	Description
Scripts and Add-ins	File	Open Fusion's Scripts and Add-Ins manager from the File menu, which works with no document open.

Share

Command	Location	Description
Get a Share Link	Share Menu	Turn on sharing and copy the public link, with the share state reported.
Change Share Settings	Share Menu	Open Fusion's share settings: download allowed, password.
Get Open on Desktop Link	Share Menu	Copy a link that opens the document in a teammate's Fusion.
Get Open in Team Link	Share Menu	Copy a link that opens the document in the Fusion Team web client.
Invite to Project	Share Menu	Open the project's Invite Members page in your browser.
Document Project Members	Share Menu	Open the project's Members page in your browser.

Dev-Notes

Developer documentation, system context, C4 diagrams, the add-in lifecycle and the command-module pattern, is in [docs/arch/architecture.md](#). Setup, tooling and debugging are in the [developer guide](#). The user-documentation style guide is [docs/dev/docs-style.md](#).

AI coding agents start at [AGENTS.md](#), which links the [codebase map](#) and the [lessons ledger](#).

Support

This add-in is developed and maintained by IMA LLC.

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The three Autodesk-sample-derived modules in `lib/ptAddInUtils`, `general_utils.py`, `event_utils.py` and `attributes_utils.py`, remain under Autodesk, Inc.'s original, permissive license terms (see each file's header), which require that their copyright notice be retained in all copies. The proprietary terms above do not apply to those Autodesk-derived portions.

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